

NETFLIX DATA ANALYSIS USING POWER BI

Power BI & Media Analytics Project Report

Domain: Entertainment, Media & OTT Analytics

1. Executive Summary

Netflix Data Analysis Using Power BI is an analytics project designed to organize, clean, and analyze content catalog, customer viewing behavior, subscription plans, and rating feedback. The project focuses on using Power BI and Excel to answer practical streaming media business questions such as identifying top-performing shows and movies, understanding customer demographic preferences, analyzing subscription engagement, and supporting regional localization and content release decisions.

The project combines data cleaning, exploratory data analysis (EDA), subscription analytics, genre popularity mapping, rating distributions, and interactive dashboard creation to establish a structured foundation for OTT business intelligence.

2. Problem Statement

Netflix is a global streaming platform specializing in movies and TV shows. As the platform expands, management needs to optimize content strategy, improve customer engagement, and maximize subscription retention while managing large volumes of show metadata, customer demographics, watch time, and rating data.

The project aims to clean and visualize this information effectively, support meaningful analytical reporting, and help management make data-driven content and subscription decisions.

3. Business Use Cases & Objectives

- **Top Content Identification:** Identify the most popular movies and TV shows on Netflix based on watch time and user ratings.
- **Target Audience Demographics:** Analyze customer demographics to understand target audience preferences across age groups.
- **Subscription Tier Impact:** Understand the impact of subscription types (Basic, Standard, Premium) on watch time and customer engagement.
- **Content Ratings & Viewer Behavior:** Explore content maturity ratings (TV-MA, PG-13, etc.) and their effect on viewing habits.
- **Optimal Release Window:** Provide insights on the best time for content release to maximize audience engagement.

- **Refined Content Strategy:** Assist Netflix in refining its acquisition strategy based on genre popularity and rating feedback.
- **Global Customer Retention:** Compare customer retention and engagement patterns across different geographical countries.

4. Dataset Overview

Dataset	Purpose & Key Variables
Netflix Shows.csv	Contains catalog metadata: show_id, type, title, director, cast, country, date_added, release_year, rating, duration, genre.
Netflix Customer Ratings.csv	Contains user viewing behavior: customer_id, show_id, customer_name, age, gender, country, subscription_type, watch_time_minutes, rating.

5. Data Architecture & Relational Structure

The project is structured around two relational datasets linked by the key field show_id:

- **Netflix Shows → Customer Ratings:** A show in the catalog links to multiple customer viewing records and ratings via show_id.
- **Customer Demographics → Watch Time:** Customers are linked to specific subscription tiers (Basic, Standard, Premium) and watch time minutes.
- **Content Ratings → Viewership:** Content maturity ratings (TV-MA, PG-13, TV-14, etc.) map directly to audience age groups.

6. Analytical Modules & Power BI Implementation

Module Code	Analytical Feature Name	Scope & Description
MOD-01	Data Consistency & Preprocessing	Identify missing directors, cast, and country fields; standardize date_added formats and age ratings.
MOD-02	Customer Demographics Analysis	Group audience into age brackets (13-18, 19-30, 31-45, 46-60, 60+) and evaluate watch time distributions.
MOD-03	Subscription Tier Watch Time	Analyze relationship between plan types (Basic, Standard, Premium) and total watch minutes to find engagement leaders.

MOD-04	Top Catalog Rankings	Identify top 10 most-watched titles based on total watch time and compare Movies vs. TV Shows consumption.
MOD-05	Genre Popularity by Country	Find the top genre per country based on viewer count and display geographic trends on a world map chart.
MOD-06	Customer Ratings Distribution	Analyze user ratings (1-5 stars) and calculate average rating per genre and geographic region.
MOD-07	Trending Content Dynamics	Identify recent additions (last 2 years) and compare average ratings of newly added vs. older content.
MOD-08	Subscription Market Share	Evaluate market share of Basic, Standard, and Premium subscription plans across global markets.
MOD-09	Customer Retention Insights	Determine single-show vs. multi-show viewer patterns and analyze if heavy watchers give higher ratings.
MOD-10	Quality Score Benchmarking	Identify highest and lowest-rated genres and compare feedback scores for TV Shows vs. Movies.
MOD-11	Cast & Director Influence	Analyze if popular directors/cast yield higher ratings and list the top 5 highest-rated directors.
MOD-12	Content Localization Strategy	Identify underperforming countries in engagement/ratings and suggest localized content strategies.
MOD-13	Optimal Release Scheduling	Analyze monthly content addition cadence and cross-reference with ratings to find optimal release periods.
MOD-14	Customer Engagement Dashboard	Build an interactive Power BI dashboard featuring watch time, ratings, and slicers (Country, Age, Tier).

MOD-15	Competitor Benchmarking Simulation	Compare Netflix content availability/ratings by country and simulate strategic platform improvements.

7. High-Value Audience Analysis

The project identifies high-value customer segments by calculating total watch minutes per subscriber, ranking users according to engagement, and identifying heavy streaming users (over 5,000 watch minutes). This analysis supports targeted retention campaigns and premium plan upgrade strategies.

8. Advanced Visual & DAX Analysis

- **Total Watch Time per Content Type:** Calculating aggregate watch minutes by joining Netflix Shows and Customer Ratings using dynamic DAX measures.
- **Top Engaged Viewers:** Identifying customers with the highest total watch time during selected release periods.
- **Director Impact Ranking:** Finding top-rated directors with the most active catalog titles and high user feedback.

9. Data Standardization & Preprocessing

The raw dataset is cleaned and standardized by handling missing values in director, cast, and country columns, converting date_added into standard date formats, and bucketing ratings into structured categories. Data preprocessing reduces inconsistencies and supports a clean Power BI data model.

10. Dashboard & Interactive Slicers

- **Demographic Filtering:** Slicers allow stakeholders to filter visuals by Age Bracket (13-18, 19-30, 31-45, 46-60, 60+).
- **Geographic & Subscription Slicers:** Interactive map and plan filters allow deep-dive analysis by Country and Subscription Tier (Basic, Standard, Premium).

11. Real-World Business Insights

- **Peak Engagement Demographics:** Which age group and subscription plan account for the highest watch hours?
- **Top Performing Genres:** What are the three most frequently watched content genres globally?

These outputs help management focus acquisition budgets, understand viewer concentration, and prioritize high-demand content categories.

12. Key Project Outcomes

- **Data Model Design:** Designed a clean, structured multi-table data model in Power BI.
- **Actionable Visual Analytics:** Developed dynamic dashboards answering critical streaming business questions.
- **Strategic Stakeholder Reports:** Delivered executive-ready reporting decks with data-backed recommendations.
- **Optimized Release Windows:** Identified optimal monthly release windows to maximize viewership and user ratings.

13. Conclusion

Netflix Data Analysis Using Power BI demonstrates how streaming data and visual analytics can transform operational platform data into useful business intelligence. By combining data cleaning, demographic grouping, subscription tier analysis, genre mapping, release timing optimization, and interactive dashboard design, the project provides a practical framework for customer, content, regional, and revenue strategy analysis.